# HUMAIRA FATHIMA N

**Data Analyst | AI Engineer**

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## PROFESSIONAL SUMMARY

Data Analyst and AI Engineer who builds Python data pipelines, predictive models and LLM applications. Hands-on with RAG, LangChain, embeddings and vector search, plus NLP, forecasting and statistical analysis. Currently delivering procurement analytics at CEGIS Foundation. Deploys AI services with FastAPI and Docker, and communicates findings through dashboards and written recommendations.

## PROFESSIONAL EXPERIENCE

**Program Associate – Data Analyst**

*CEGIS Foundation* | Chennai | Jan 2026 – Present

- Built Selenium, Playwright (stealth) and BeautifulSoup pipelines to collect and structure public procurement tender data from Tamil Nadu, Odisha and Uttar Pradesh.
- Developed a CatBoost model estimating which bidders are most likely to win or lose a tender from historical bidding data.
- Analysed bidder patterns to flag potential cartelisation and collusion for review.
- Automated data collection and reporting with Google APIs, reducing manual handling.
- Cleaned and validated large, inconsistent datasets for analysis and reporting.

## PROJECTS

**RAG-Powered PDF Chatbot with Citation Support** | Python, LangChain, Hugging Face, Vector DB, FastAPI, Docker

- End-to-end RAG pipeline (chunking, embeddings, semantic retrieval, LLM synthesis) with citation-backed answers, served via REST API.

**Multiclass Fish Image Classification** | Python, CNN, ResNet50, Deep Learning, Image Processing

- Built a CNN-based multiclass fish image classifier using the ResNet50 architecture, with image preprocessing before training and evaluation.

**PowerPulse: Household Energy Forecasting** | Python, ARIMA, Machine Learning, Feature Engineering

- Time-series forecasting of smart-meter consumption combining ARIMA and ML models; $R^2$ of 0.89 after feature engineering.

**Nifty 50 Stock Analysis Platform** | Python, Pandas, SQL, Power BI, Streamlit

- Wrangling, statistical analysis and feature engineering on Nifty 50 data, with an interactive Streamlit dashboard.

## CORE SKILLS

**Languages:** Python, SQL

**GenAI & LLMs:** RAG, LangChain, Hugging Face, Embeddings, Vector DBs, Prompt Design, Ollama

**ML & NLP:** NLP, CatBoost, Predictive Modelling, Classification, Feature Engineering, TensorFlow, Keras, CNN

**Analytics:** Statistical Analysis, ARIMA Forecasting, Data Cleaning & Validation, Power BI, Streamlit

**Engineering:** FastAPI, REST APIs, Docker, Git, Selenium, Playwright, Google APIs

## SOFT SKILLS

- Communication & data storytelling
- Stakeholder engagement
- Cross-functional collaboration
- Mentoring
- Strategic thinking

## EDUCATION

**MCA (Pursuing)**
Manipal University Online

**BCA**
CTTE College for Women, Chennai | Aug 2022 – Apr 2025

## CERTIFICATIONS

**Advanced AI & Machine Learning**
IIT Madras (via GUVI) | Aug 2024 – Feb 2025

- Coursework in machine learning, deep learning, NLP and applied AI.
- Applied these methods in projects that I extended into the RAG and forecasting work above.